

Main Topic

Fault-Tolerant Embedded Edge AI Systems

Subtopic

Multimodal Fault Detection and Reliable Activation in Assistive Sensing Systems

Tentative Title

Fault-Tolerant Multimodal Presence Verification for Privacy-Preserving Assistive Systems on ESP32-S3

Abstract

Modern assistive and smart-environment systems increasingly rely on edge AI deployed on embedded platforms for real-time perception and interaction. However, such systems are highly susceptible to faults arising from sensor noise, environmental variability, timing uncertainties, and limited computational resources. Systems that rely on a single sensing modality are prone to incorrect activation decisions, leading to false positives (unsafe or unnecessary activation) and false negatives (missed assistance), both of which degrade system reliability and user trust.

This project proposes the design and evaluation of a fault-tolerant multimodal perception system implemented on an ESP32-S3 platform, integrating a camera module and a 24GHz mmWave radar sensor. The system focuses on reliable human presence verification in a room-scale environment. The mmWave sensor provides robust detection of human presence independent of lighting conditions, while the camera provides spatial and visual confirmation. These heterogeneous sensing modalities are used redundantly to detect inconsistencies and identify potential faults.

The system incorporates fault detection mechanisms such as temporal redundancy, cross-sensor consistency checking, and anomaly detection in sensor outputs. Detected inconsistencies are further classified into transient and persistent faults. Based on this classification, the system employs adaptive fault-handling strategies. A two-tier safe-state mechanism is introduced: a soft safe state for temporary inconsistencies that triggers re-validation, and a hard safe state for persistent faults that blocks system activation, isolates unreliable inputs, and optionally notifies the user.

The proposed approach prioritizes minimizing unsafe activations (false positives) while maintaining acceptable responsiveness to valid user presence. The system operates under a conservative activation policy supported by multimodal verification, ensuring that assistive actions are only triggered under high-confidence conditions.

The effectiveness of the system will be evaluated through controlled fault injection, including sensor noise, dropped data, and timing delays. Performance will be compared against a baseline non-fault-tolerant implementation using metrics such as false activation rate, missed detection rate, fault detection coverage, and system latency.

This work demonstrates how classical fault tolerance principles (redundancy, system-level diagnosis, error detection, graceful degradation, etc.) can be applied to modern edge AI systems to improve reliability, safety, and trustworthiness in assistive applications.

Note on Scope

The proposed work represents a focused subsystem of a broader ongoing project involving a multimodal assistive perception and interaction platform on ESP32-S3 (for CPRE 5750). The larger project includes additional capabilities such as gesture recognition, audio interaction, and dynamic user tracking using servos. The current work isolates and studies the fault tolerance aspects of presence detection and system activation, enabling a deeper and more rigorous exploration of reliability mechanisms within a manageable scope.